

Project Proposal

Project Topic:	<i>Development and Operationalization of Data Science Solutions – Predictive Maintenance of Aircraft Engines</i>
Project abbreviation:	
Institute/ Chair/ Research Group:	<i>Institut für Technische und Betriebliche Informationssysteme</i>
Advisor(s):	<i>Prof. Dr. Klaus Turowski, M. Sc. Christian Haertel</i>
Preferred group size:	<i>4</i>
Desired project period:	<i>5 months</i>
Required/Desired knowledge:	Python, Data Analysis, Data Engineering, Machine Learning, Software Engineering, Technical Writing
Is any external affiliation involved (e.g., industrial partner, affiliated institute)? yes ☐ No ☒ Which one(s)?	
<u>Project Description:</u> Data Science (DS) is a discipline that aims to extract knowledge from data. This process often involves the application of machine learning (ML). DS teams often face managerial and technical hurdles during project execution. Therefore, engineering practices as prescribed by the MLOps paradigm are increasingly applied to DS to help with the development, deployment, and monitoring of ML models. The goal of the project is the end-to-end realization (i.e., from conceptualization to the utilization of the solution) of a DS use case. For this purpose, the team should follow a provided MLOps-based DS process model to manage the project. The DS problem to be addressed originates from the transportation sector. Aircraft engines are subject to wear over time, making them unsafe for further use after a certain point in time. Therefore, to implement data-driven predictive maintenance, an ML-based solution should be developed that predicts engine breakdowns based on sensor readings. Dataset: https://www.kaggle.com/datasets/mhadani/predictive-maintenance-aircraft-engine/data Project deliverables: – <i>Implementation of an end-to-end Data Science use case based on the provided problem, following an MLOps-based process model</i> – <i>Comprehensive and detailed documentation of the implementation according to the process model, including conducted activities and encountered challenges</i> – <i>Code base</i> <i>Tasks:</i>	

- *Familiarization with reference literature on Data Science and MLOps, including the MLOps-based process model*
- *Conceptualization of the business problem and project planning (Business Understanding)*
- *Dataset exploration and preparation (Data Collection, Exploration, and Preparation)*
- *Selecting, training, and evaluating ML models (Modeling)*
- *Evaluation of the project status concerning business goals (Evaluation)*
- *Preparing and executing the deployment of the ML-powered system, including testing (Deployment)*
- *Demonstration of the use, monitoring, and maintenance processes for the prototypical system (Utilization)*
- *Detailed and scientifically robust documentation of all managerial and technical implementation steps to ensure reproducibility, including descriptions and architectural diagrams*
- *Critical reflection on the project, including the used techniques, technologies, and the MLOps-based process model*

Related Literature:

- *Kreuzberger, Dominik; Kühl, Niklas; Hirschl, Sebastian (2023): Machine Learning Operations (MLOps): Overview, Definition, and Architecture. In: IEEE Access 11, S. 31866–31879. DOI: 10.1109/ACCESS.2023.3262138.*
- *Material on the MLOps-based DS process model: <https://github.com/c-haertel/MLOps-based-Data-Science-Process-Model/tree/main>*